

Functional Annotation Report: *coaX* (Q88QQ0) — Type III Pantothenate Kinase, *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

The target is confirmed and **unambiguous**. A direct 2026 query of UniProtKB for pantothenate kinases (EC 2.7.1.33) in *Pseudomonas putida* KT2440 (taxid 160488) returns exactly **one** entry: **coaX / PP_0438 / Q88QQ0**, "Type III pantothenate kinase (PanK-III)." The gene symbol, EC number, organism, protein family (type III pantothenate kinase), and domain complement (ATPase_NBD IPR043129; Type_III_PanK IPR004619; Pfam PF03309 Pan_kinase) all match the literature for the CoaX/PanK-III family. No conflicting gene with the same symbol was encountered. All claims below apply to this specific protein.

1. Summary (Answer to the Research Question)

coaX (PP_0438, UniProt Q88QQ0) encodes a **type III pantothenate kinase (PanK-III/CoaX; EC 2.7.1.33)**, the enzyme that catalyzes the **first and committed step of coenzyme A (CoA) biosynthesis**: phosphorylation of the vitamin **(R)-pantothenate** using **ATP** to yield **(R)-4'-phosphopantothenate + ADP** (Rhea:16373). It is a soluble **cytoplasmic, homodimeric** enzyme that requires a **monovalent cation (K⁺ or NH₄⁺)** for activity and belongs to the **ASKHA (acetate/sugar kinase-Hsp70-actin) structural superfamily**. Unlike the *E. coli*-type PanK-I, PanK-III is **not feedback-inhibited by CoA**, has an **unusually high K_m for ATP**, and **cannot use pantothenamide analogues** as substrates. Because *coaX* is the **sole pantothenate kinase gene** in *P. putida* KT2440, it is the non-redundant gatekeeper of the CoA pool and is expected to be **essential**.

2. Primary Function: Reaction and Substrate Specificity

Reaction catalyzed (EC 2.7.1.33, Rhea:16373):



This is **step 1 of the universal 5-step CoA biosynthetic pathway** ("CoA from (R)-pantothenate: step 1/5") and is the first committed, generally rate-controlling reaction of the pathway (UniProt Q88QQ0, HAMAP-Rule MF_01274; Yang et al. 2006,

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Yang et al. 2008,

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Substrate specificity. The phosphoacceptor is the primary hydroxyl of **pantothenate (vitamin B5)**; the phosphate donor is **Mg-ATP**. Biophysical characterization of a Gram-negative PanK-III homolog (*Acinetobacter baumannii*) found **strong pantothenate affinity ($K_d \approx 1.2 \times 10^{-8}$ M)** and only **moderate ATP affinity ($K_d \approx 3.7 \times 10^{-3}$ M)** (Singla et al. 2021,

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— consistent with the family-wide signature of an **unusually high K_m for ATP** (Yang et al. 2008,

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What PanK-III does NOT do. PanK-III is **resistant to feedback inhibition by CoA and its thioesters**, and is **unable to phosphorylate N-alkylpantothenamide analogues** — so it does not convert those antimetabolites into toxic CoA analogues in vivo (Yang et al. 2008,

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These properties sharply distinguish it from the *E. coli* CoaA (type I) enzyme, in which CoA feedback inhibition is the principal regulatory valve of the pathway (Yang et al. 2006,

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3. Cofactor and Catalytic Mechanism

- **Monovalent-cation dependence.** CoA/UniProt annotation specifies a required **monovalent cation, either NH_4^+ or K^+** (UniProt Q88QQ0, HAMAP-Rule MF_01274). This K^+/NH_4^+ activation is a defining biochemical hallmark of PanK-III that distinguishes it from PanK-I and PanK-II.
- **Fold and superfamily.** The first PanK-III crystal structure (*Thermotoga maritima*, 2.0 Å) established that type III PanKs adopt the **ASKHA (acetate and sugar kinase/Hsp70/actin) fold** and retain the superfamily's conserved active-site motifs (Yang et al. 2006,

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This is consistent with the ATPase nucleotide-binding-domain annotation (IPR043129) for Q88QQ0.

- **Catalytic residues.** Mutagenesis of **three highly conserved active-site aspartates** identified them as **critical for catalysis** and explained the enzyme's lack of CoA feedback inhibition (Yang et al. 2006,

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In Q88QQ0 the annotated catalytic active site is **residue Asp102**, with a P-loop/nucleotide-binding motif (residues 6–13) and additional substrate/metal-binding residues (93, 100–103, 122, 125, 181) (UniProt Q88QQ0).

- **Substrate-bound structures.** Binary complexes with **pantothenate** (substrate) and **phosphopantothenate** (product), and a **ternary complex with pantothenate + ADP**, together with isothermal titration calorimetry, defined the substrate-binding mode and thermodynamics of PanK-III catalysis (Yang et al. 2008,

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4. Quaternary Structure and Subcellular Localization

- **Quaternary structure: homodimer** (UniProt Q88QQ0, HAMAP-Rule MF_01274). The mature protein is 249 aa.
- **Localization: cytoplasm** (GO:0005737; UniProt-SubCell). CoA biosynthesis is a soluble cytosolic process, and the enzyme acts on intracellular pantothenate and Mg-ATP. There are

no signal peptides or membrane-spanning features; the enzyme carries out its function in the bacterial cytosol.

5. Pathway Context and Biological Process

CoaX initiates the conserved five-step conversion of pantothenate to CoA:

1. **Pantothenate → 4'-phosphopantothenate** (pantothenate kinase, **CoaX/PanK-III** — this enzyme)
2. 4'-phosphopantothenate + cysteine → 4'-phosphopantothenoylcysteine (PPC synthetase, CoaB)
3. → 4'-phosphopantetheine (PPC decarboxylase, CoaC)
4. → dephospho-CoA (phosphopantetheine adenylyltransferase, CoaD/PPAT)
5. → CoA (dephospho-CoA kinase, CoaE)

CoA and its 4'-phosphopantetheine moiety are indispensable cofactors for acyl-group transfer across central metabolism — fatty-acid synthesis/oxidation, the TCA cycle, and hundreds of acyl-CoA-dependent reactions — and CoA additionally serves as a **low-molecular-weight thiol** in bacterial redox/antioxidant defence (Gout 2019,).

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By setting flux into this pathway, CoaX governs the supply of the cell's CoA pool. Notably, because PanK-III is CoA-insensitive, the **feedback regulation of the CoA pool in PanK-III organisms is exerted at downstream steps** (e.g., phosphopantetheine adenylyltransferase/PPAT, which is product-inhibited by CoA), rather than at the kinase step (Miller et al. 2007,).

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Genomic context (*P. putida* KT2440)

KEGG assigns PP_0438 to **KO K03525** (type III pantothenate kinase) within pathway **ppu00770 "Pantothenate and CoA biosynthesis,"** at genome positions **531,036–531,785 (forward strand)**. The gene is **not part of a dedicated CoA-biosynthesis operon**; instead its start codon overlaps (~10 bp) the stop of the immediately upstream same-strand gene *birA* (PP_0437, bifunctional biotin-[acetyl-CoA-carboxylase] ligase/biotin operon repressor), and its 3' end

overlaps PP_0439 (conserved hypothetical). This tight overlap suggests **probable co-transcription/translational coupling with *birA***, functionally linking two cofactor-supply systems that both feed acyl/lipid metabolism (CoA for acyl-group chemistry; biotinylation of acetyl-CoA carboxylase for fatty-acid synthesis). Immediately downstream lie highly expressed housekeeping genes (EF-Tu/*tuf*, *secE*, *nusG*), consistent with constitutive expression of an essential biosynthetic enzyme. No second (type I) *coaA* gene is present anywhere in the genome. (*Bioinformatic inference from gene neighborhood; the operon has not been experimentally mapped.*)

6. Essentiality, Distribution, and Druggability

- **Distribution:** PanK-III is **widely distributed across bacteria** and is the **only PanK in many species, including *Pseudomonas aeruginosa*, *Helicobacter pylori*, and *Bordetella pertussis*** (Yang et al. 2006,

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- **Sole PanK in *P. putida*:** the UniProt inventory confirms *coaX*/PP_0438 is the **only** pantothenate kinase gene in *P. putida* KT2440, making it non-redundant.
- **Essentiality is context-dependent.** In *Mycobacterium tuberculosis*, which encodes **both** a type I *coaA* and a type III *coaX*, *coaA* could only be deleted with an extra gene copy while ***coaX* was dispensable** (no growth defect in vitro, in macrophages, or in mice) — CoaA is the essential PanK there; by contrast, **CoaX is essential in *Bacillus anthracis***, where it is the sole PanK (Awasthy et al. 2010,

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Applying this logic, *coaX* is **expected to be essential in *P. putida*** because it is the organism's only PanK.

- **Drug target:** PanK-III is an attractive **selective antibacterial target** — it is essential (in sole-PanK organisms) and **structurally divergent from human PanK ($\approx 28\%$ identity)** (Singla et al. 2021,

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The broader CoA pathway is a well-validated antimicrobial target space (Butman et al. 2020,

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7. Supported and Refuted Hypotheses

Supported: - H1: *coaX* encodes a functional type III pantothenate kinase catalyzing pantothenate + ATP → 4'-phosphopantothenate + ADP. **Supported** (UniProt catalytic activity; family structural/enzymatic studies

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- H2: The enzyme is CoA-insensitive with high K_m for ATP and cannot use pantothenamides.

Supported

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homolog kinetics

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- H3: It adopts an ASKHA fold with catalytic aspartates; functions as a K^+/NH_4^+ -dependent cytoplasmic homodimer. **Supported**

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UniProt Q88QQ0). - H4: *coaX* is the sole PanK and therefore essential in *P. putida*. **Supported by inference** (UniProt gene inventory + sole-PanK essentiality argument,

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not yet directly tested genetically in *P. putida*.

Refuted / Qualified: - The blanket claim "*coaX* is always essential" is **refuted** — it is dispensable where a type I CoaA is co-encoded (M. tuberculosis,

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Essentiality depends on being the sole PanK.

8. Limitations and Future Directions

- **No *P. putida*-specific enzymology or knockout** was found in the literature; the functional assignment rests on (i) UniProt/HAMAP curation of Q88QQ0, (ii) strong structural/biochemical studies of PanK-III orthologs (*T. maritima*, *A. baumannii*), and (iii) the

essentiality logic. Direct biochemical characterization and a conditional knockout of PP_0438 in *P. putida* KT2440 would confirm essentiality and native kinetic parameters.

- The K^+/NH_4^+ **cofactor** requirement for the *P. putida* enzyme is annotation-based (HAMAP-Rule MF_01274); species-specific verification would be valuable.
- Future work: solve/validate the *P. putida* CoaX structure (or AlphaFold model) to confirm Asp102 and the dimer interface, and evaluate CoaX-selective inhibitors as anti-Pseudomonad leads.

Key References

- Yang K, Eyobo Y, Brand LA, et al. *J Bacteriol* 2006;

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— first PanK-III crystal structure; ASKHA fold; catalytic aspartates; CoA-insensitive; sole PanK in *Pseudomonas*, *H. pylori*, *B. pertussis*.

- Yang K, Strauss E, Huerta C, Zhang H. *Biochemistry* 2008;

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— substrate/product/ternary complex structures; high K_m for ATP; CoA-resistant; no pantothenamide use.

- Awasthy D, et al. *Microbiology* 2010;

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— *coaA* vs *coaX* essentiality in *M. tuberculosis*; CoaX essential in *B. anthracis*.

- Singla M, et al. 2021;

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— *A. baumannii* PanK-III biophysics; substrate affinities; drug-target rationale; ~28% identity to human PanK.

- Miller JR, et al. *Biochemistry* 2007;

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— PPAT as the CoA-feedback-regulated step.

- Gout I. 2019;

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— CoA as protective thiol in bacterial antioxidant defence.

- Butman HS, et al. 2020;

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— CoA biosynthesis as antibacterial target space.

- UniProtKB **Q88QQ0** (HAMAP-Rule MF_01274) — reaction, cofactor, pathway, homodimer, cytoplasm, catalytic/binding residues.

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